YASHWANTH DEVAVARAPU

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Education

KL University, Andhra Pradesh, India — Aug 2021 – May 2025

B.Tech. in Computer Science and Engineering (Hons.) | **CGPA: 9.6/10** | Ranked among the **Top 5% students**

- **Honors & Awards:** KL Scholar (2021–2025); Smart India Hackathon (SIH) 2022 – National Finalist and Runner-up.
- **Research Recognition:** Google DeepMind APAC Research Symposium Student (2026); Finalist, Meta Open Source AI Hackathon.
- **Teaching Assistant, Java Full Stack:** Selected to mentor students in Java, Spring, Hibernate, and microservices; led hands-on guidance and debugging sessions, helping students resolve implementation challenges and develop independent problem-solving skills.

Research Experience

Research Collaborator – Stanford University (MS&E) — Jun 2026 – Present

- With **Prof. Ross Shachter**, investigated LLM reliability in decision analysis, uncovering a response-ordering artifact in which answer-first prompts produced values ranging from $60 to $82.35, while reasoning-first prompts consistently produced $76.00; demonstrated how evaluation design can distort conclusions about LLM reasoning.
- Developed a code-generation evaluation pipeline achieving exact solutions on **128/128 decision problems**, where in-context reasoning failed; showed that error feedback improved gpt-4o from 10.2 to 23.0 out of 29 tests, while leaving a performance gap.

Research Collaborator – Harvard Medical School / Massachusetts General Hospital — Mar 2026 – Present

- Co-led **Dr. Xiangrui (Taylor) Zeng**'s research on unpaired 3D brain MRI generation; developed a wavelet diffusion model, reducing reconstruction MSE from 0.088 to **0.0022** over 3M+ training steps, with segmentation conditioning achieving **over 0.92 SSIM**.
- Currently developing generative models for anatomically conditioned and longitudinal brain MRI prediction; implementing MSC label conditioning and consistent evaluation pipelines to assess anatomical fidelity and prediction quality across time points.

Research Collaborator – Carnegie Mellon University, Prof. Min Xu's Lab — Jan 2026 – Present

- **MIRA (CVPR 2026 MMRAgI Workshop):** First-authored *Through the Looking Glass*, a benchmark for mirror-image reasoning in vision-language models; constructed an evaluation of 11 models across 713 images, 3,529 QA pairs, and 10 scenarios, revealing 39–48% model accuracy against 94% for human participants across 54K+ queries.
- **Multi-LoRA Composition (ICLR 2027 submission):** Evaluated routing and attention mechanisms for diffusion-adapter composition, extending experiments to three-adapter settings and multi-seed ablations. Contributed Cross-Adapter Attention Conditioning (CLAC), improving identity consistency from 0.839 to **0.915**. (Research collaborator: **Dr. Xingjian Li**)
- **TRACE / APIShift:** Currently investigating agentic systems for network troubleshooting and structured tool execution, focusing on BGP policy faults, silent network failures, and action-schema adherence across model scales.

Research Collaborator – With Pierre Ntakirutimana, Research Associate, CMU Africa — 2026 – Present

- **TabuLM:** Co-developed a morphology-aware Kinyarwanda tabular language model using structural embeddings and table-aware attention; contributed to pretraining on 172 govt tables (˜35K cells) and the TabQA-kin benchmark, achieving **62.0% exact match**.
- **KinyaEmbed:** Developed a multilingual embedding model and four-stage contrastive-learning curriculum; contributed to model and benchmark evaluation, achieving a **0.7298 Spearman correlation** on SemRel2024-rw, outperforming mE5-large by 20.9%.

Undergraduate Research Assistant – KL University — Aug 2022 – Mar 2025

- **Credit Card Fraud Detection:** First-authored an **IEEE CONIT 2024** paper with **Dr. K. Ashesh**, developing an outlier-based fraud detection framework using Isolation Forest, Local Outlier Factor, and One-Class SVM to identify anomalous transaction patterns.
- **Multimodal Emotion Detection:** Led the development of a customer emotion analysis system integrating VGG16-based facial, CNN–LSTM speech, and fine-tuned BERT text models; achieved **92% validation accuracy** through weighted late fusion.

Work Experience

Software Engineer – Barclays, Pune, India — July 2025 – Present

- Architected and developed an **8-microservice internal developer platform** using Java 21, Spring Boot, PostgreSQL, and AWS, integrating vulnerability management, FinOps, database operations, and infrastructure automation into a centralized platform; scaled adoption to **100+ applications** across **50+ engineering teams**.
- Built **LangGraph-based AI agents** with MCP tool integrations to automate security scans, CVE detection, Jira ticket creation, and reporting; reduced manual triage and received the **Director's Recognition Award** for improving automation reliability.
- Developed AI-driven container vulnerability remediation and validation pipelines spanning **15,000+ services** and **4,000+ applications**, reducing remediation time from several days to **under one hour**.
- Designed and developed specialized AI agents for **Java 8-to-21 application modernization**, automating dependency analysis, code transformation, compatibility checks, testing, and validation workflows to reduce migration timelines from weeks to days.
- Designed a hybrid **AWS SES–Keycloak authorization model** with dynamic role provisioning and self-service onboarding for **50+ teams**; developed multi-account FinOps capabilities for cost analytics, anomaly detection, forecasting, and savings recommendations, alongside GitLab migration and CI/CD onboarding for **100+ repositories**.

Software Engineering Intern – Barclays, Pune, India — June 2024 – July 2024

- Developed a full-stack cache monitoring platform using **Spring Boot, Thymeleaf, and Apache Ignite** to track SDS and ESM cache metrics across distributed systems, providing centralized visibility into cache activity and performance.
- Implemented anomaly detection to identify unusual cache fluctuations and monitor performance variations across distributed environments, enabling faster identification of potential cache-related issues and improving operational visibility.
- Built automated email alerts for cache fluctuations and optimized data retrieval workflows, improving retrieval speed by **30%**.

Publications

- **MVEB: Massive Video Embedding Benchmark** – EMNLP 2026
  Introduced a **23-task benchmark** evaluating 33 video embedding models, revealing modality-dependent performance across classification, retrieval, clustering, and video question answering.
- **EdgeMedBench: Benchmarking Small Language Models for Medical QA** – IC4AI 2026
  Evaluated **5 small language models** across 45 configurations, identifying an INT4 deployment achieving 50% average accuracy with 1.18 GB VRAM and 190 ms inference latency.
- **ML-Based SDN DDoS Detection: Machine Learning-Enabled IDS** – CIMA 2026
  Developed an ML-based intrusion detection framework for SDN environments, evaluating multiple classifiers on CICIDS-2017 to identify DDoS, DoS, brute-force, and other network attacks.
- **Reproducible, Cross-Platform Study of Order-Book Data Structures in a Low-Latency Matching Engine** – arXiv 2026
  Compared **3 order-book implementations** using reproducible cross-platform benchmarks, demonstrating 1.6–2.4× higher throughput than a third-party engine and identifying OS- and allocator-dependent performance differences.
- **Physics-Informed Neural Network Analysis of Rotational Effects on Oscillatory MHD Flow** – EPJP Journal
  Investigated rotational effects on steady and oscillatory MHD flows using physics-informed neural networks, incorporating governing equations and boundary constraints for flow-field reconstruction.
- **Research Acknowledgment – AICO: Feature Significance Tests for Supervised Learning – PNAS 2026**
  Acknowledged by **Prof. Kay Giesecke (Stanford University)** and **Chartsiri Jirachotkulthorn** for providing research comments and feedback on their work on feature significance testing for supervised learning.
- **Technical Reviewer – IEM-HEALS 2026**
  Served as a technical reviewer for the International Conference on Advancing Science and Technologies in Health Science, evaluating research submissions for technical quality and relevance.

Selected Projects

Bitsmith – Open-Source LLM Quantization & GPU Kernel Engineering

- Engineered an open-source quantization framework from first principles, implementing **INT8, group-wise INT4, FP8 (E4M3/E5M2), ternary encoding, and 2:4 structured sparsity** with actual sub-byte packing into uint8 buffers; integrated SmoothQuant, AWQ, and block-sequential GPTQ with activation ordering for calibration-aware compression across 3 language models.
- Developed a fused INT4 dequantization–matrix multiplication Triton kernel, performing register-level dequantization to eliminate intermediate weight unpacking; achieved up to **1.33× speedup over FP16 cuBLAS** and 5–24× over the reference INT4 path on selected GPU layers. Reduced SmolLM2-135M INT4 perplexity degradation from 28.9% to **12.7%** using GPTQ; validated with 209 CPU tests at 97% coverage and 85 GPU tests.
  Tech: Python, PyTorch, Triton, NumPy, CUDA, Hugging Face Transformers.

PageLLM – LLM Inference Engine & High-Performance Serving

- Built an LLM inference engine from scratch using Python and NumPy, implementing a Llama-style decoder with **RMSNorm, RoPE, GQA, and SwiGLU**; engineered a block-allocated **Paged KV cache** with block tables, reference counting, copy-on-write, and LRU-based prefix caching, reducing KV-cache waste from 61.7% to **7.1%** and supporting 3.8× more concurrent sequences under the evaluated workload.
- Implemented a continuous-batching scheduler with token-budget-based chunked prefill, batched decode attention, and speculative decoding with rejection sampling; achieved **1.21× higher throughput**, **2.06× lower p99 inter-token latency**, and **2.46× faster prefix-cached execution**. Built an OpenAI-compatible HTTP/SSE serving layer, concurrent load-testing infrastructure, and automated validation with **164 tests at 96% coverage**.
  Tech: Python, NumPy, Transformer Architecture, KV Caching, HTTP, SSE, Docker.

KafkaCore Lab – Apache Kafka Internals & Performance Engineering

- Building a Java-based benchmarking and regression-testing framework for Apache Kafka to systematically evaluate **throughput, p95/p99 latency, and resource utilization** under configurable workloads and varying message loads.
- Investigating broker internals, KRaft controller behavior, partition replication, and failure recovery; developing automated fault-injection scenarios and regression tests to validate message integrity, broker recovery, and overall cluster resilience.
  Tech: Java, Apache Kafka, KRaft, Gradle, JUnit, Linux, Docker.

ChatQuill – Real-Time Chat Application

- Developed a real-time chat application using **WebSocket-based bidirectional communication**, enabling instant message exchange between connected users and reducing communication latency by **50%** compared to conventional request-response interactions.
- Implemented **STOMP-based messaging** for message broadcasting, subscription management, and real-time user presence tracking, supporting reliable communication across multiple active user sessions.
  Tech: Java, Spring Boot, WebSockets, STOMP, Thymeleaf, HTML, CSS.

Technical Skills

- **Languages & Programming**: Java, Python, C++, C, SQL, Bash
- **AI & Machine Learning**: LLMs, VLMs, Multimodal Learning, Diffusion Models, LoRA, Model Fine-Tuning, PyTorch, Hugging Face Transformers, CLIP
- **Agentic AI & Evaluation**: Multi-Agent Systems, Agent Orchestration, MCP, Tool Calling, RAG, LangGraph, CrewAI, Model Benchmarking, AI Evaluation, Decision Analysis
- **Data & Infrastructure**: Apache Spark, Redis, Vector Databases, FAISS, ChromaDB, AWS Bedrock, SageMaker, OpenTelemetry
- **Engineering & DevOps**: Linux, Git, GitLab, Docker, Kubernetes, CI/CD, Microservices, Distributed Systems